

80Y



80Y

12K

80Y

12K

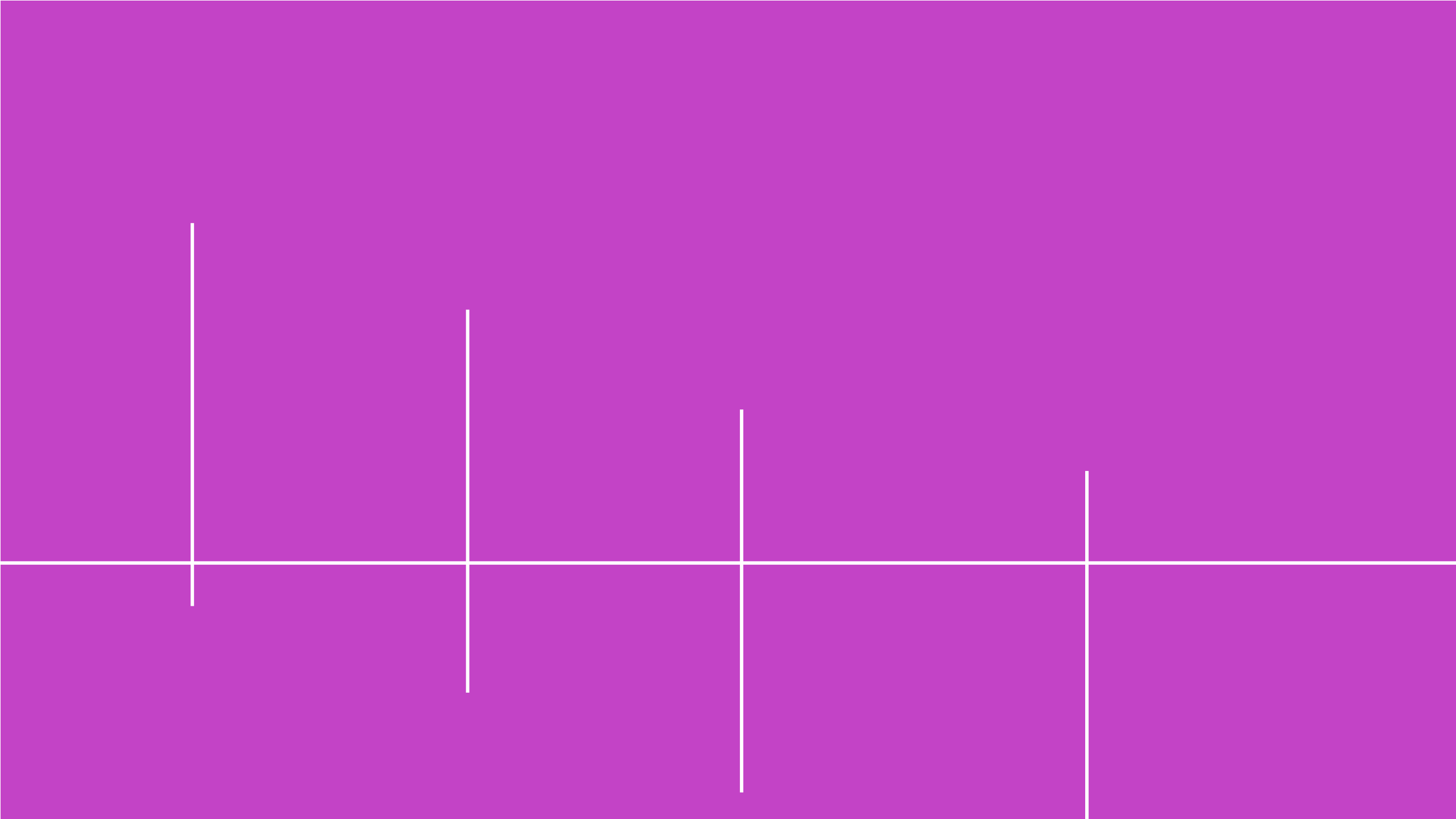
12.5B

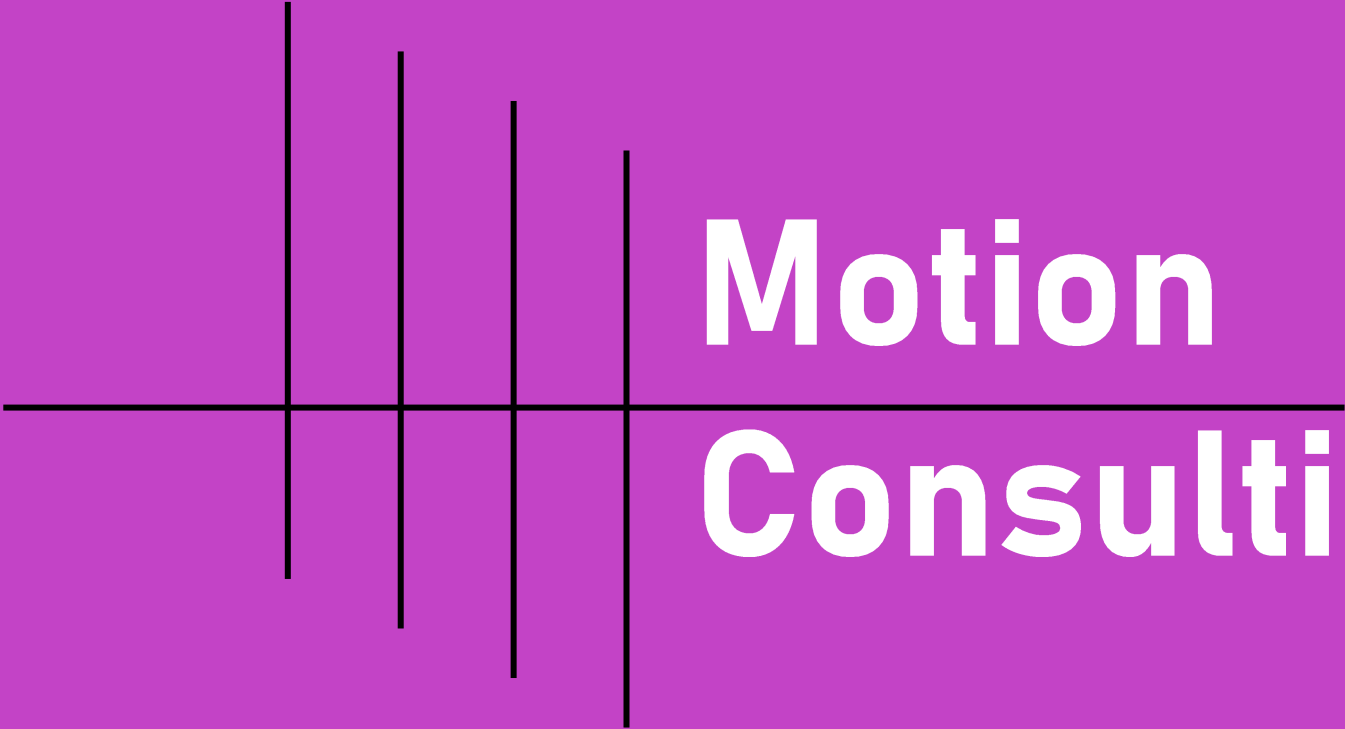
80Y

12K

12.5B

3,500 KM





Motion Consulting

Prepared for

Prairie Oil



Diya Arora



Kritin Kumar



Daisy Dyck

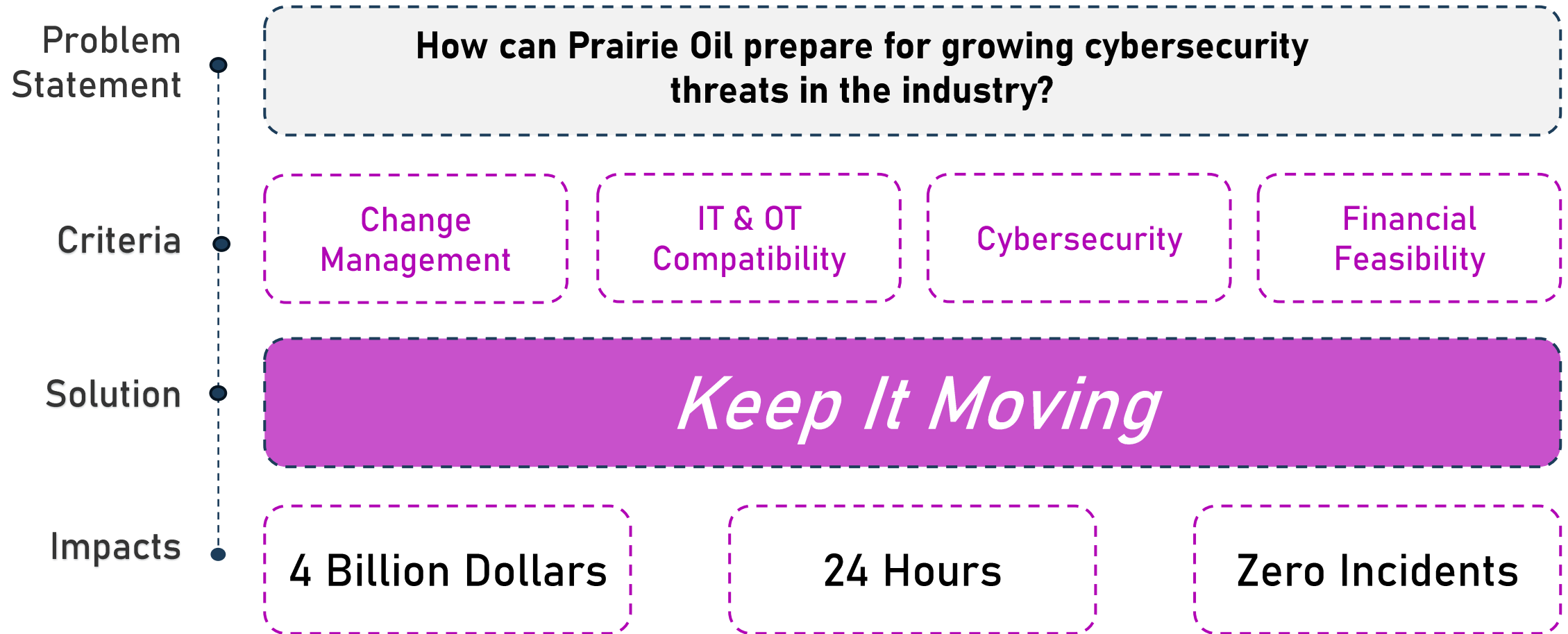
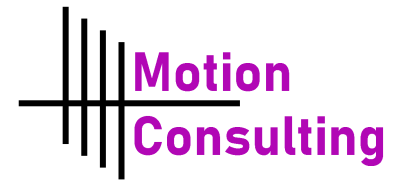


Christian Arteaga

**How can Prairie Oil prepare
for growing cybersecurity
threats in the industry?**

Executive Summary

How can we help?



Analysis

Recommendation

Implementation

Impacts

Risks

Personal Impacts

How does this affect the individuals?

Vivek



Age: 36 (IT Lead)

- Has worked with Prairie Oil for 5 years now
- Dreads audits asking him, "*How is your system secure?*"
- Wants a system that allows patching without risking downtime
- Struggles to balance security with day-to-day IT duties

Analysis

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Vivek



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- Dreads audits asking him, "*How is your system secure?*"
- Wants a system that allows patching without risking downtime
- Struggles to balance security with day-to-day IT duties

Kaitlynn



Age: 46 (Pipeline Supervisor)

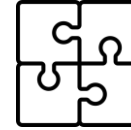
- Has worked at Prairie Oil for 17 years
 - Pipeline supervisor for a large sector of pipelines across Alberta
 - "If it isn't broken, don't fix it"
- Pain Point:**
- Still relies on the legacy Honeywell TDC 3000, but fears change

People, Process, Technology



Current State

- Stressed by change
- Facing nuisance due to legacy systems
- Unprepared for cyber threats



Desired State

Analysis

Recommendation

Implementation

Impacts

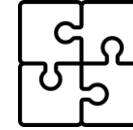
Risks

People, Process, Technology



Current State

- Stressed by change
- Facing nuisance due to legacy systems
- Unprepared for cyber threats



Desired State

- Prepared for growing cybersecurity risks
- Reduced workload due to outdated systems



Analysis

Recommendation

Implementation

Impacts

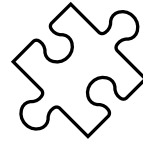
Risks

People, Process, Technology



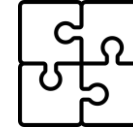
Current State

- Stressed by change
- Facing nuisance due to legacy systems
- Unprepared for cyber threats



Key Enabler

Updated operations with a
change management
implementation plan



Desired State

- Prepared for growing cybersecurity risks
- Reduced workload due to outdated systems



Analysis

Recommendation

Implementation

Impacts

Risks



Current State



Desired State

Key Takeaway

Key Enabler

Employees will be vulnerable in the event of any system changes

- Stressed by change
- Facing nuisance due to legacy systems
- Unprepared for cyber threats

Operated operations with change management implementation plan

Prepared for growing cybersecurity risks
Reduced workload due to outdated systems

Criteria:
Change Management

Analysis

Recommendation

Implementation

Impacts

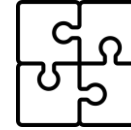
Risks

People, Process, Technology

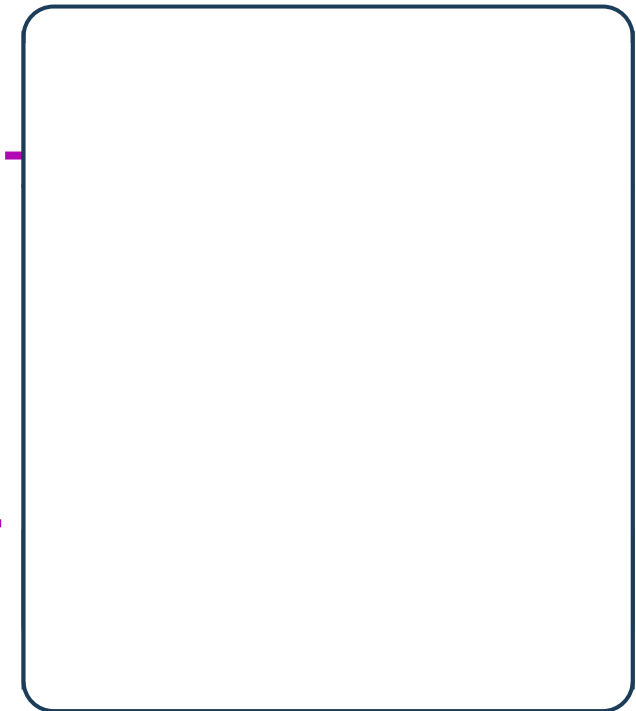


Current State

- 3 Canadian refineries with siloed data, lacking security measures
- Data analyzed locally and sent to the head office via microwave
- IT and OT integration relies on ad-hoc solutions



Desired State



Analysis

Recommendation

Implementation

Impacts

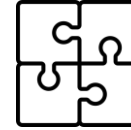
Risks

People, Process, Technology



Current State

- 3 Canadian refineries with siloed data, lacking security measures
- Data analyzed locally and sent to the head office via microwave
- IT and OT integration relies on ad-hoc solutions



Desired State

- Centralized data storage and management
- Independent Integration between IT and OT systems
- Create “perfect fit” rather than “functional”

Analysis

Recommendation

Implementation

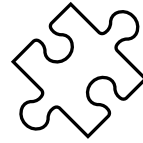
Impacts

Risks



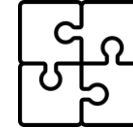
Current State

- 3 Canadian refineries with siloed data, lacking security measures
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- IT and OT integration relies on ad-hoc solutions



Key Enabler

Completing migration to new operating system and finding a compatible data storage system



Desired State

- Centralized data storage and management
- Independent Integration between IT and OT systems
- Create “perfect fit” rather than “functional”



Current State

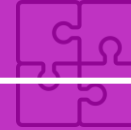
- 3 Canadian refineries with siloed data, lacking security measures
- Data analyzed locally and sent to the head office via microwave
- IT and OT integration relies on ad-hoc solutions



Key Takeaway

Key Enabler

Integrated independence between
IT & OT must be stronger



Desired State

- Centralized data storage and management
- Solid integration between IT and OT systems
- Create “perfect fit” rather than “functional”

Criteria:

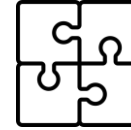
IT & OT Compatibility

People, Process, Technology



Current State

- Using Honeywell TDC 3000, which in 2025 will be 41 years old
- Partial migration to Siemens PCS 7, leaving fragmented security practices



Desired State

Analysis

Recommendation

Implementation

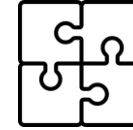
Impacts

Risks



Current State

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Desired State

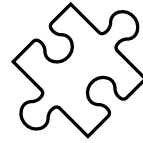
- No longer reliant on ancient operating systems
- No more stagnant migration
- Secure systems with cybersecurity mindset

People, Process, Technology



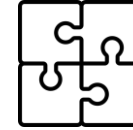
Current State

- Using legacy Honeywell TDC 3000, which is over 40 years old
- Partial migration to Siemens PCS 7, leaving fragmented security practices



Key Enabler

Prairie Oil needs a new system that allows for critical processes to be protected



Desired State

- No longer reliant on ancient operating systems
- No more stagnant migration
- Secure systems with cybersecurity mindset

Analysis

Recommendation

Implementation

Impacts

Risks



Current State

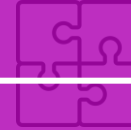
- Using Honeywell TDC 3000, which in 2025 will be 40 years old
- Partial migration to Siemens PCS 7, leaving fragmented security practices



Key Takeaway

Key Enabler

The new system must provide up-to-date cyber threat security measures



Desired State

- No longer reliant on ancient operating systems
- No more stagnant migration
- Secure systems with cybersecurity mindset

Criteria:
Cybersecurity



Analysis

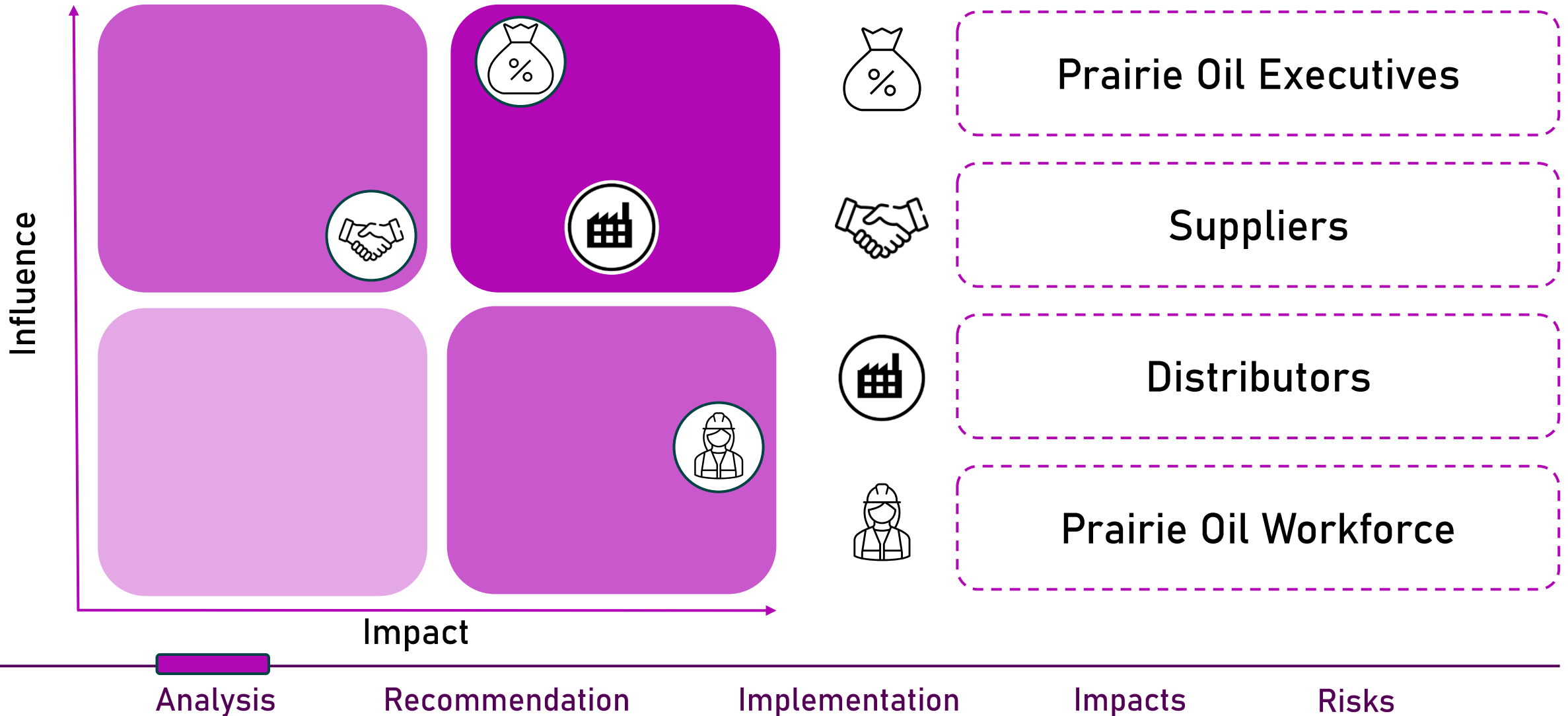
Recommendation

Implementation

Impacts

Risks

Stakeholder Analysis

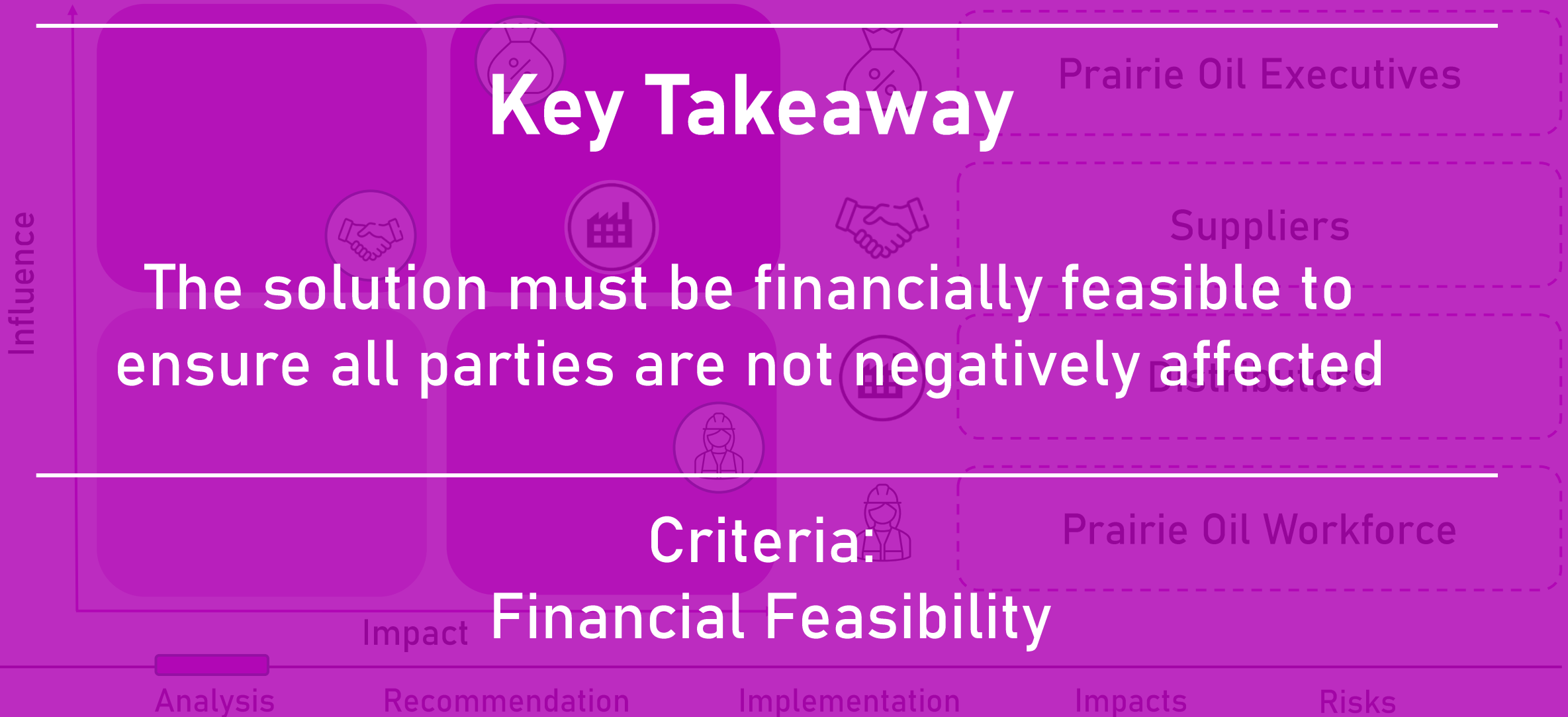


Key Takeaway

The solution must be financially feasible to ensure all parties are not negatively affected

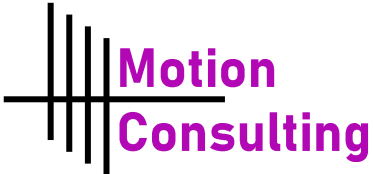
Criteria:

Financial Feasibility



Alternatives

Assessing potential strategies against criteria



	Change Management	IT & OT Compatibility	Cybersecurity	Financial Feasibility
Increase cybersecurity without updating OT	●	●	●	●
AI implementation	●	●	●	●
Migrate to Honeywell Experion	●	●	●	●
Hybrid Internal Build	●	●	●	●
Siemens PCS 7	●	●	●	●

Analysis

Recommendation

Implementation

Impacts

Risks

Alternatives

Assessing potential strategies against criteria

	Change Management	IT & OT Compatibility	Cybersecurity	Financial Feasibility
Increase cybersecurity without updating OT	●	●	●	●
AI implementation	●	●	●	●
Migrate to Honeywell Experion	●	●	●	●
Hybrid Internal Build	●	●	●	●
Siemens PCS 7	●	●	●	●

Analysis

Recommendation

Implementation

Impacts

Risks

Keep It Moving

1

Data Integration and Migration

- Move from siloed data into AWS
- Implement backup and encryption

2

Internal Structural Changes

- Ensure OT and IT compatibility
- Network segmentation

Keep It Moving

Goal:

Prioritize cyber safety with

minimal downtime

1

Data Integration and Migration

- Move from siloed data into AWS
- Implement backup and encryption

2

Internal Structural Changes

- Ensure OT and IT compatibility
- Network segmentation

Data Integration and Migration

Phase 1

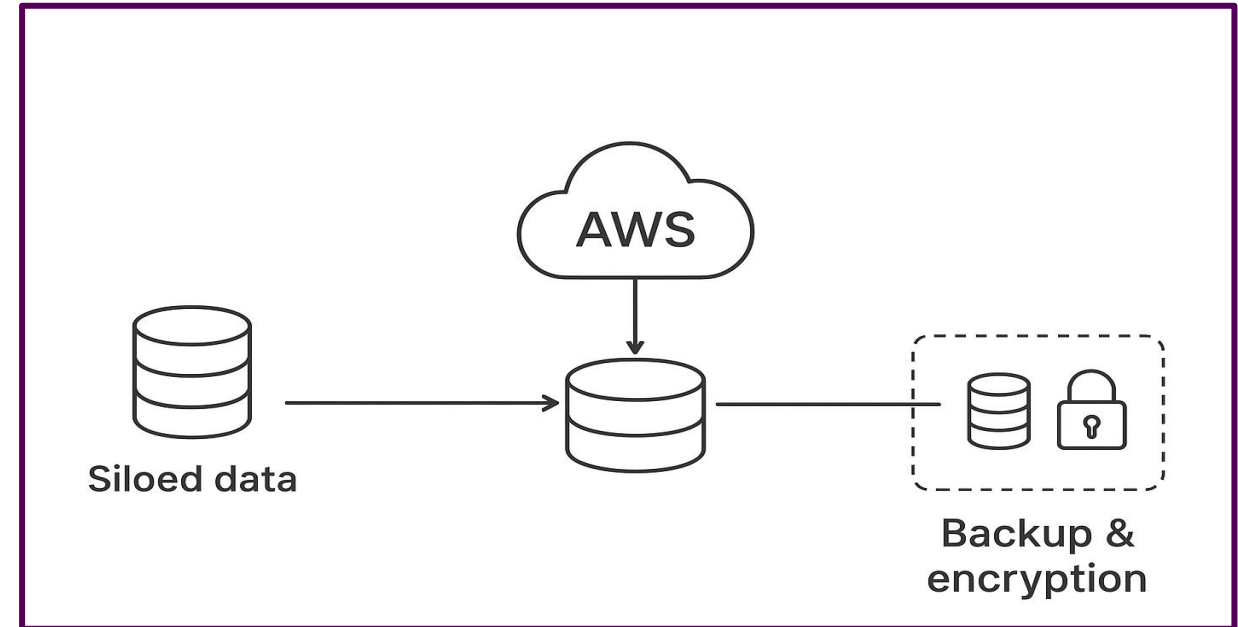
Transition to centralized data lake (AWS)

Key Features:

AWS Identity Access and Management:
User/role permissions for access

AWS Key Management Service:
Encryption Key Management

AWS Audit Manager:
Automates evidence collection for audits



Internal Structural Changes

Phase 2

Migration to Siemens PCS 7

Key Features:

Integrated DCS Platform

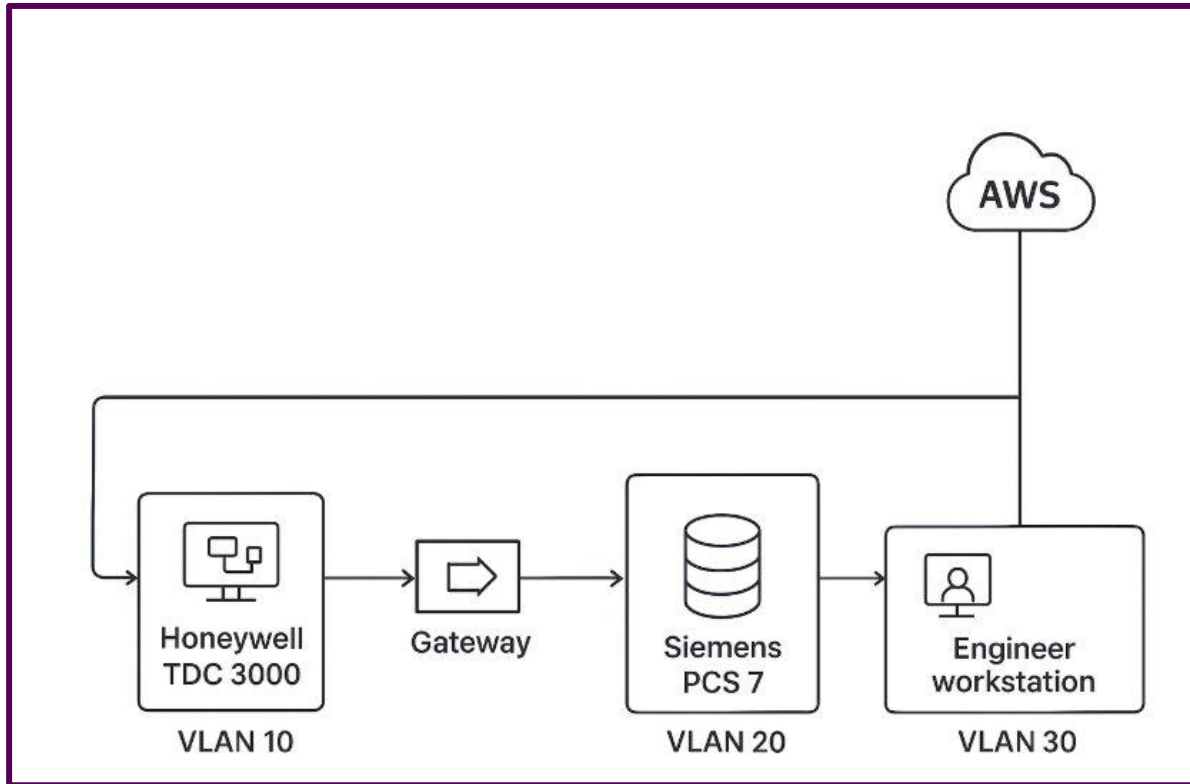
Process control, safety systems, batch control, and asset management in one environment.

Scalability and Flexibility

Handles anything from a single compressor station up to enterprise-wide pipeline operations across thousands of I/O points.

Cybersecurity Standard

Built to align with IEC 62443 industrial cybersecurity standard.



OT-IT Compatibility

Phase 3

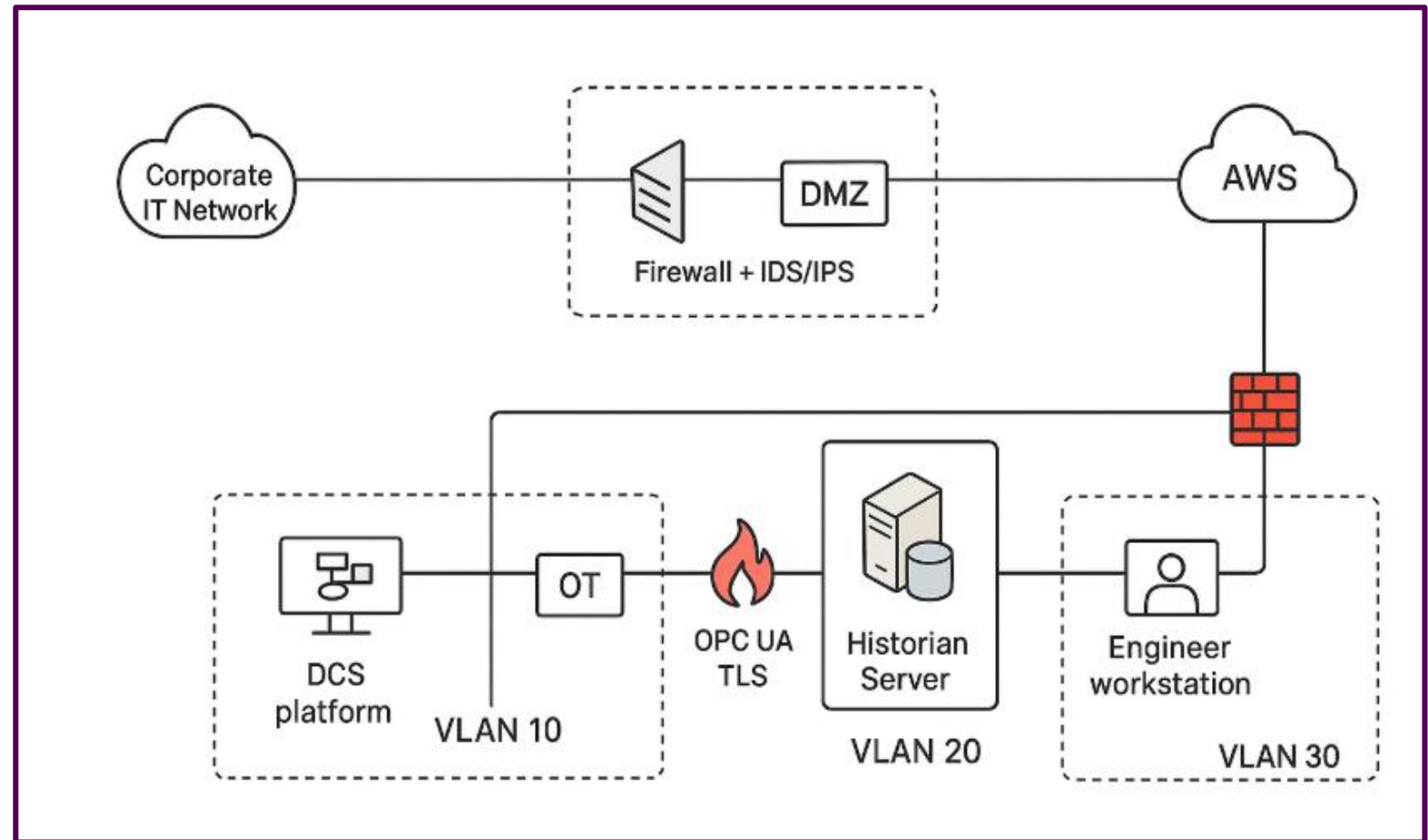
Key Features:

Siemens PCS 7 (OT Backbone)

Role-based access, system hardening, patch management, secure communications, lifecycle support (15-20 years), and scalability.

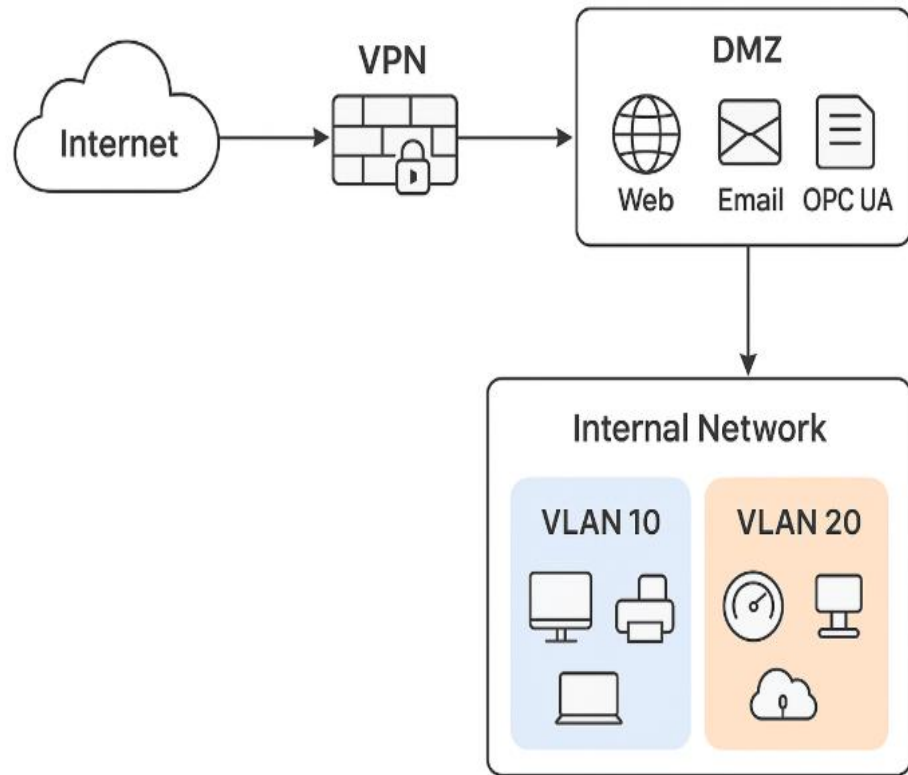
AWS Security (IT Layer)

Cybersecurity and compliance, cloud-based replicas of control center HMIs for disaster recovery, extended visibility across both OT and IT.



Network Segmentation

Phase 4



Enhance network security, control access, and protect critical systems

Key Features:

VPN Encrypted Remote Access:
Safe connection of external users to the internet.

DMZ Buffer Zone:
Hosts public-facing services while keeping internal systems protected.

VLAN Segmentation of Internal Network:
Limits lateral movement of threats.

Change Management - ADKAR

Ensuring everyone is involved.

Awareness

"The current system works fine,
why do we need to change it?"

Growing security threats and aging
equipment make this change necessary.

Is this
possible??

Why
change??



Vivek



Kaitlynn

Analysis

Recommendation

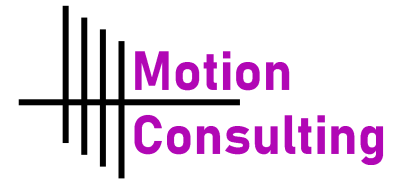
Implementation

Impacts

Risks

Change Management - ADKAR

Ensuring everyone is involved.



Awareness

"The current system works fine, why do we need to change it?"

Growing security threats and aging equipment make this change necessary.

Desire

"Why should I have to learn how to use this new technology?"

The updated system will reduce workload and shutdown time.

Analysis

Recommendation

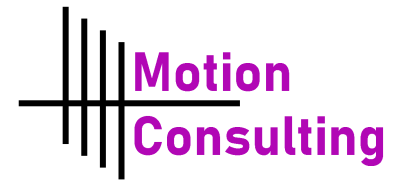
Implementation

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"Why should I have to learn how to use this new technology?"

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Knowledge

"Is there some way I can get a refresher on the new system?"

Employees will be thoroughly trained in new system prior to operational launch.

Analysis

Recommendation

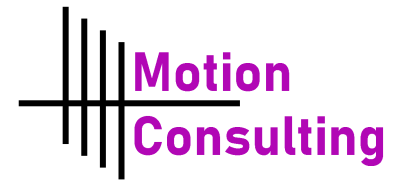
Implementation

Impacts

Risks

Change Management - ADKAR

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Employees will be thoroughly trained in new system prior to operational launch.

Ability

"I feel like some of the MFA features are redundant."

Employees can provide feedback and solutions to the minor inconveniences.

Analysis

Recommendation

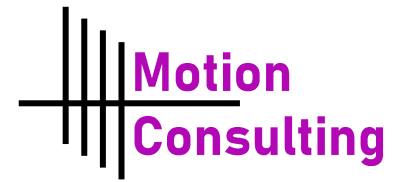
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Change Management - ADKAR

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Knowledge

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Employees will be thoroughly trained in new system prior to operational launch.

Ability

"I feel like some of the MFA features are redundant."

Employees can provide feedback and solutions to the minor inconveniences.

Reinforcement

"Will we ever go back to the old system?"

The new system is sustainable for Prairie Oil in the long-term.

Analysis

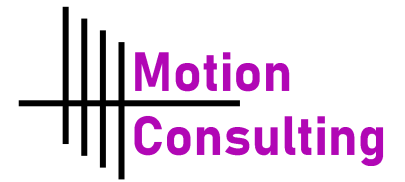
Recommendation

Implementation

Impacts

Risks

Case Studies



Proof of previous implementation



RoviSys' oil and gas company IT/OT Integration

Challenged with ensuring data quality and normalization across nearly 100 facilities, the development of standard interfaces and data quality procedures was required to ensure rapid scale-out of the solution.

Migration of a SCADA system to IaaS clouds
– A Case Study
Journal Of Cloud Computing

DTE Calvert City

Using gradual planning and an emphasis on reduced downtime, DTE Energy switched from the obsolete APACS DCS to a full Siemens PCS 7 system



Case Study

Moving the SCADA systems to an Infrastructure as a Service (IaaS) cloud allowed for: cheaper deployments, system redundancy support, and increased uptime.

Analysis

Recommendation

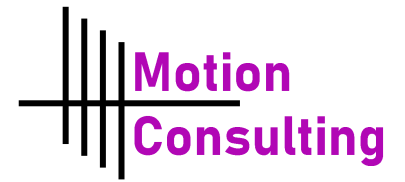
Implementation

Impacts

Risks

Immediate Implementation

How can we get started?



Tomorrow's Action Points

Kick off an organization-wide awareness campaign to show the need for change

Start AWS Cloud Transmission Process

Analysis

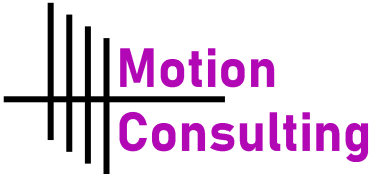
Recommendation

Implementation

Impacts

Risks

Timeline



Key tasks associated with the success of the implementation

	Short Term				Medium Term				Long Term	
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Y3	Y4
Data Migration and Integration										
Implement ADKAR method for change management										
Zero Trust baseline, AWS Cloud rollout										
Deploy Splunk SIEM (IT logs)										
DMZ, VLAN and VPN implementation										
Internal Structural Change										
OT Modernization: Honeywell to Siemens PCS 7										
Unified SOC operational (IT/OT Visibility)										
Network Segmentation and SIEM Tuning										

Future Considerations

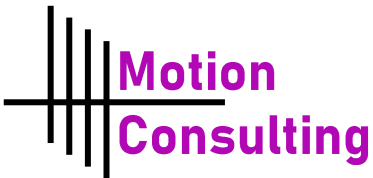
How can we keep it going?

Year 5

- Review and reassess AWS data storage
- Monitor organization adaptation to new processes

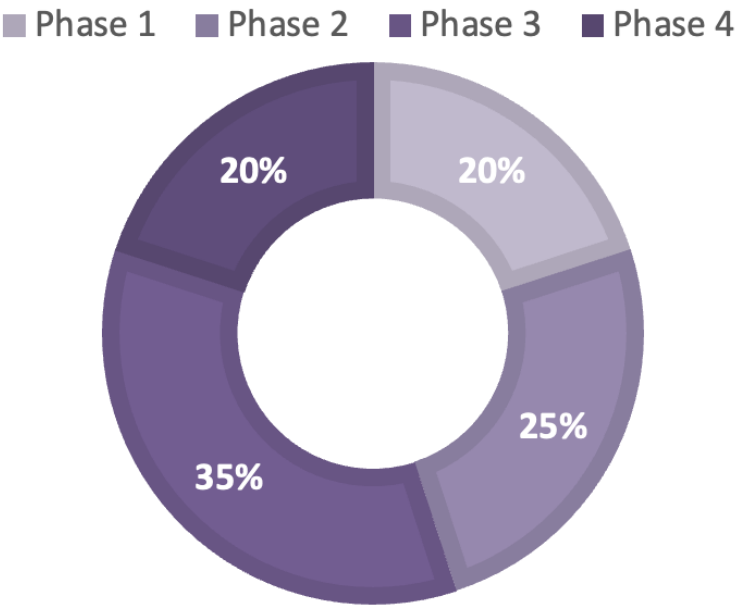
- Industry Benchmarking
- Consider AI/ML introduction

Financials



Phase 1	Cost (Million \$)
ADKAR Change Implementation	15
Workforce Training & Certifications	10
Zero Trust Baseline MFA, AWS	20
Partner Compliance Audits	15
Phase 1 Total	60

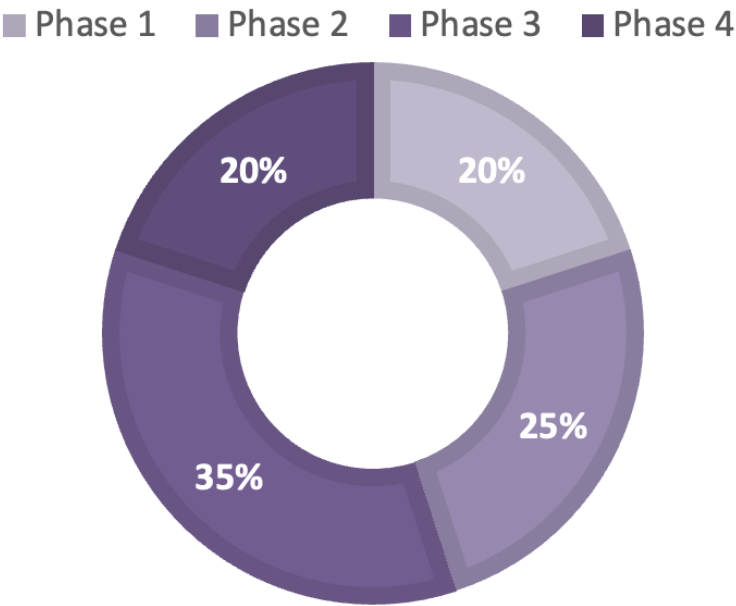
PRAIRIE OIL CYBERSECURITY
BUDGET ALLOCATION
(\$300M)



Financials

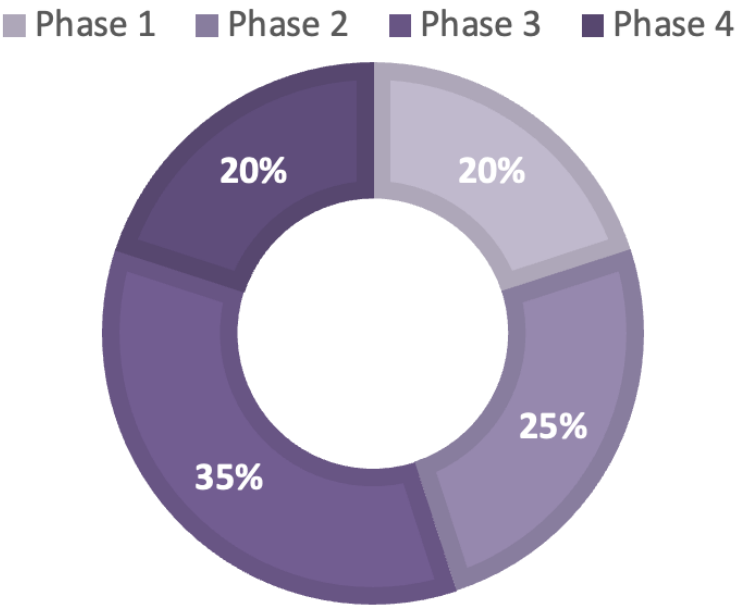
Phase 2	Cost (Million \$)
Splunk SIEM Licenses & Cloud Hosting (AWS)	40
SOC Tools & Threat Intel Feeds	20
Integration & Deployment Services	15
Phase 2 Total	75

PRAIRIE OIL CYBERSECURITY
BUDGET ALLOCATION
(\$300M)



Phase 3	Cost (Million \$)
Honeywell TDC 3000 → Siemens PCS 7 Migration	70
OT Firewalls, IDS, Secure Workstations	20
Network Segmentation & Controllers	15
Phase 3 Total	105

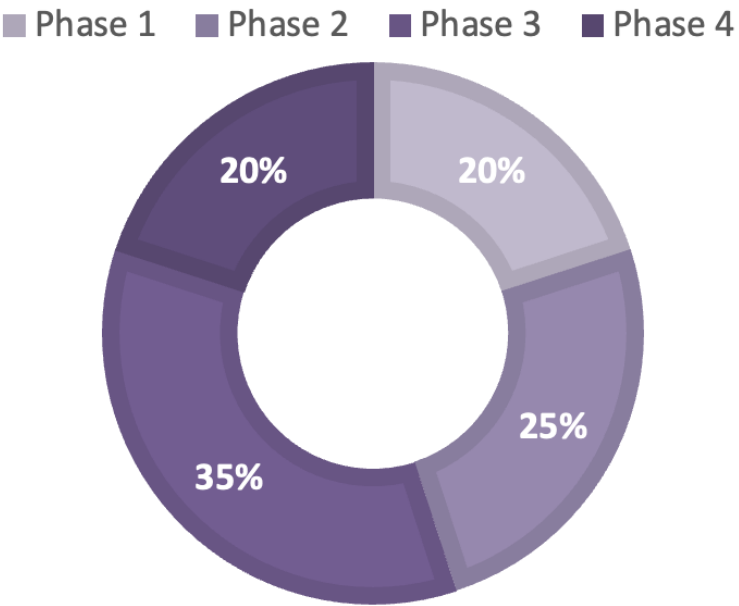
PRAIRIE OIL CYBERSECURITY
BUDGET ALLOCATION
(\$300M)



Financials

Phase 4	Cost (Million \$)
Unified SOC Staffing & Operations	25
Immutable Backups & Disaster Recovery	15
Partner/Regulator Joint Exercises	10
Continuous Improvement & Refresh Training	10
Phase 4 Total	60

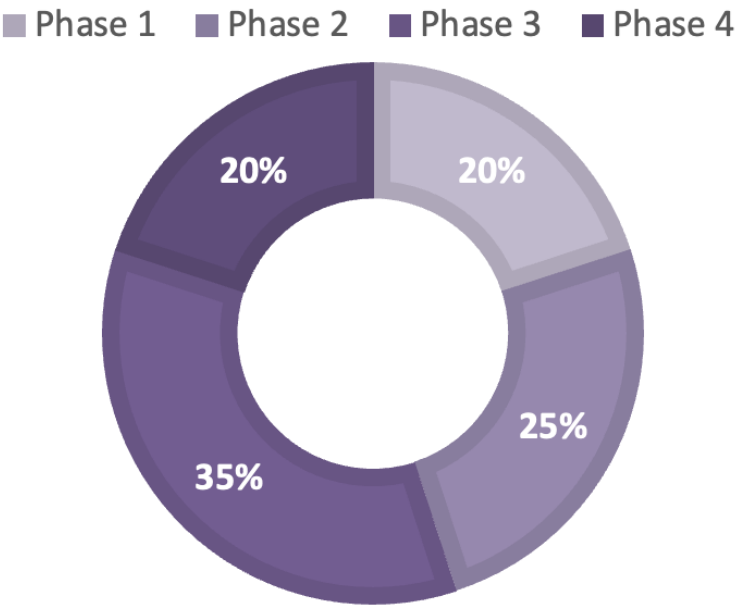
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BUDGET ALLOCATION
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Financials

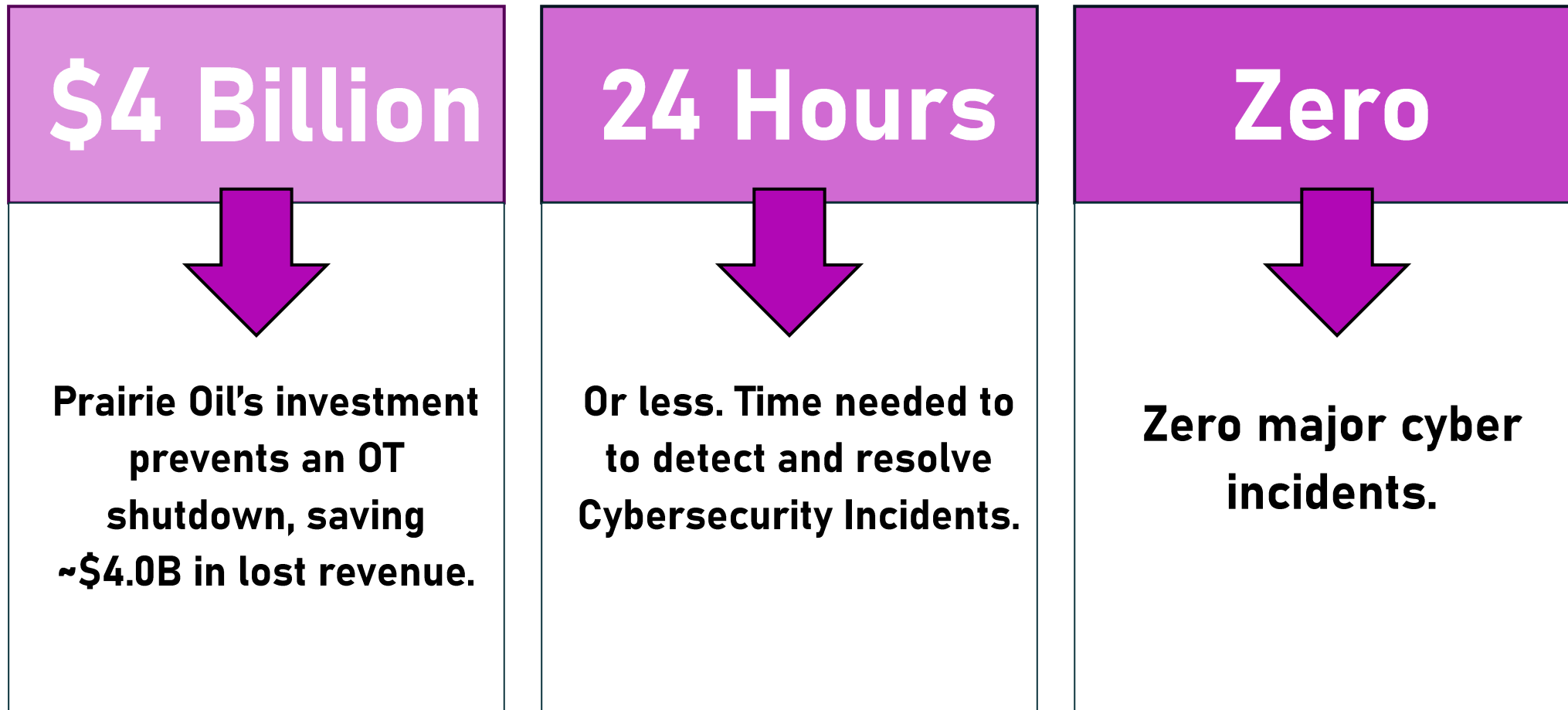
Phases	Cost (Million \$)
Phase 1	60
Phase 2	75
Phase 3	105
Phase 4	60
Total	300

PRAIRIE OIL CYBERSECURITY
BUDGET ALLOCATION
(\$300M)



Key Performance Indicators

Assessing the performance of the implementation



Analysis

Recommendation

Implementation

Impacts

Risks

Personal Impacts

Back to Vivek & Kaitlynn

Vivek



IT Lead

- Structured patch schedule that can align with plant operations, no more risk of downtime
- No longer dreads audits, knows exactly what his system is capable of because he was a part of the solution
- No more security tasks interfering with day-to-day IT duties

Kaitlynn



Pipeline Supervisor

- Lighter workload in unexpected ways, fewer nuisance alarms, intuitive technology
- Slowly regaining confidence in new operating systems
- Knew how to test, control and manage the system before it went up; her fingerprints were in the design

Analysis

Recommendation

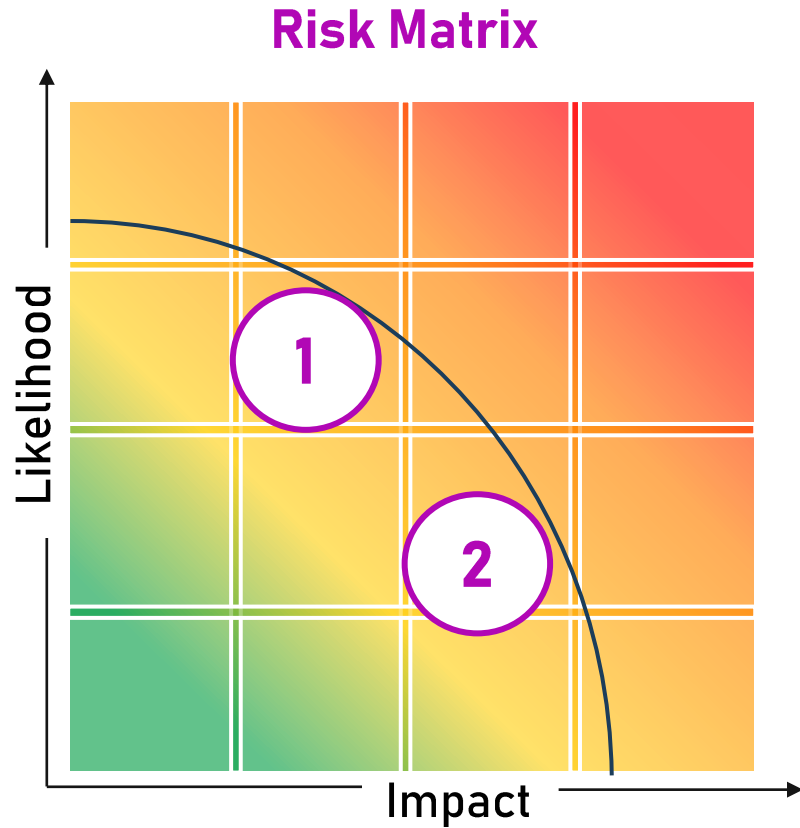
Implementation

Impacts

Risks

Risks and Mitigations

Proactive Control



Risk

1

Resistance to Change

2

Downtime During Transition

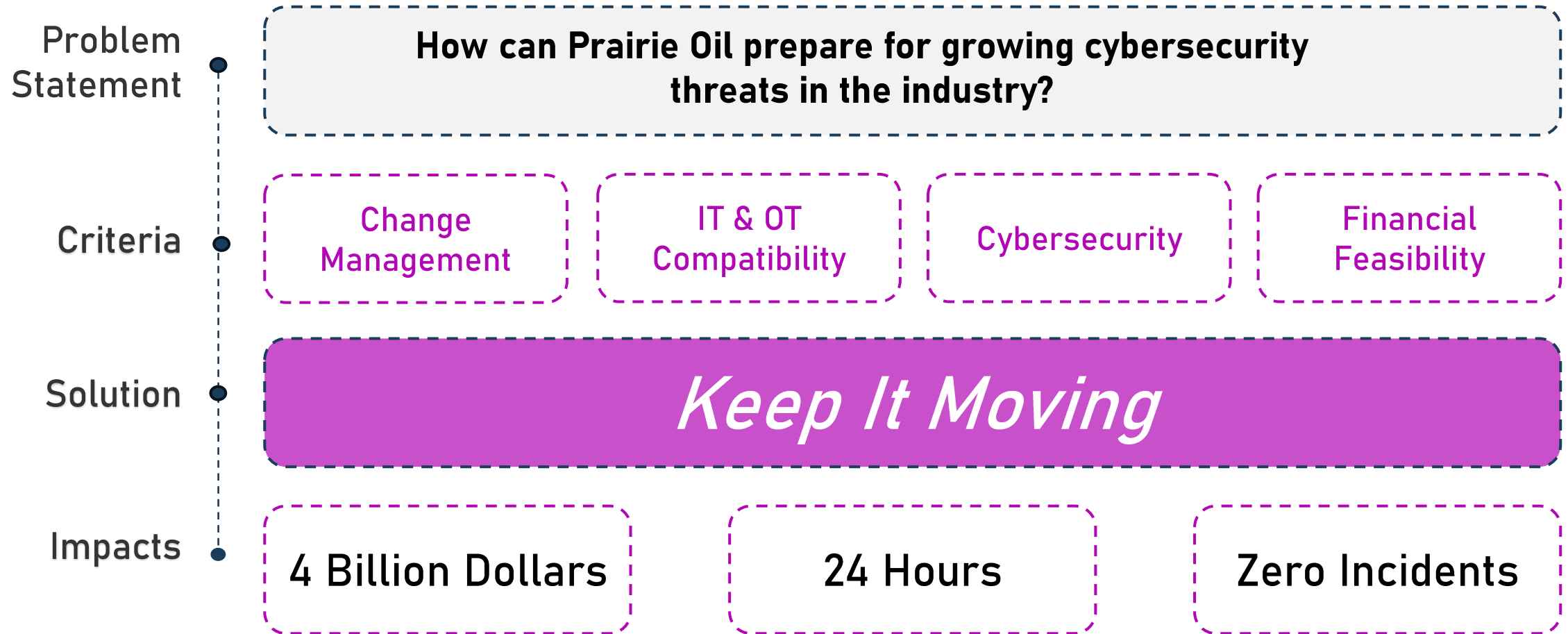
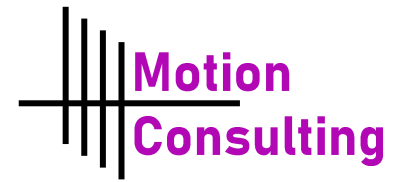
Mitigation

Ensure employees are a part of the process, implementing the ADKAR method.

Patching, segmentation, and system migrations done in phases. OT dual system during installation.

Executive Summary

How did we help?



Analysis

Recommendation

Implementation

Impacts

Risks

Q&A

Prepared for Prairie Oil



Diya Arora



Kritin Kumar



Daisy Dyck



Christian Arteaga

Appendix A

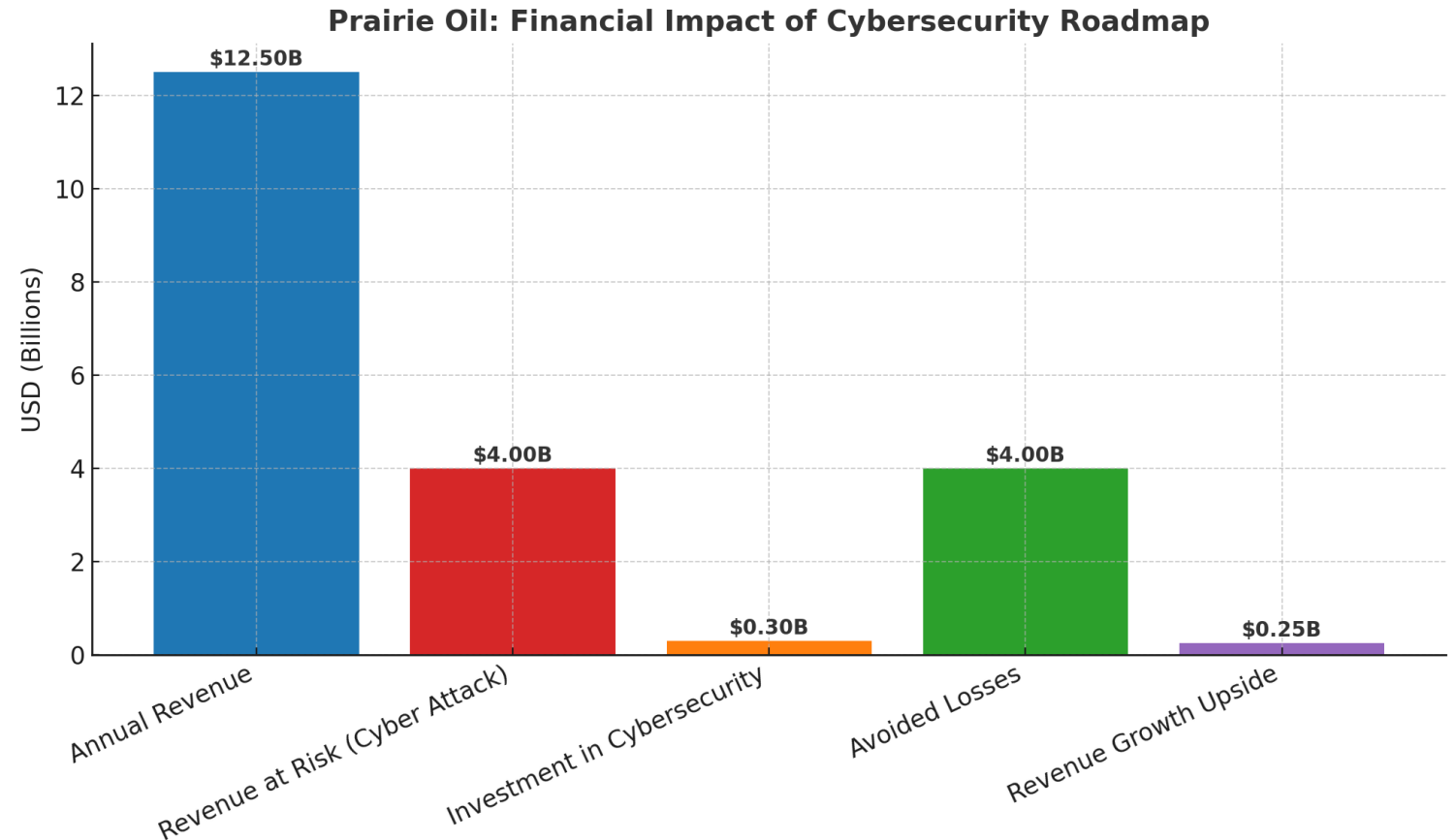
Annual Revenue: \$12.5B

Revenue at Risk (from a major cyberattack): ~\$4.0B

Cybersecurity Investment: \$0.3B
(5 years)

Avoided Losses: ~\$4.0B (Complete shutdown prevented, Shell Canada)

Revenue Growth Upside: ~\$0.25B
(contracts, uptime gains, trust benefits)



Analysis

Recommendation

Implementation

Impacts

Risks